ERRORS AND DATA TREATMENT IN QUANTITATIVE ANALYSIS

SCHOOL OF PHARMACY, TONGJI MEDICAL COLLEGE, HUST

Chp.2

01 Some important terms

02 Systematic Errors in Chemical Analysis

03 Random Errors in Chemical Analysis

04 Statistical Data Treatment and Evaluation

1.1 Percentage of stated content



Percentage of stated content

$$\% = \frac{\textit{The drug content measured}}{\textit{The drug content shown on the box}} \times 100\%$$





1.1 Percentage of stated content

A batch of paracetamol tablets are stated to contain **500 mg** of paracetamol per tablet; Percentages of stated content for the repeat analysis of paracetamol in the tablets:



***Student 1*:** 497.5, 499.5, 501, 497, 502.5

***Student 2*:** 478, 480.5, 476, 475.5, 480.5

***Student 3*:** 467.5, 491.5, 462.5, 512.5, 488

***Student 4*:** 472, 501, 522.5, 487, 510.5

***Student 1*:** 99.5%, 99.9%, 100.2%, 99.4%, 100.5%

***Student 2*:** 95.6%, 96.1%, 95.2%, 95.1%, 96.1%

***Student 3*:** 93.5%, 98.3%, 92.5%, 102.5%, 97.6%

***Student 4*:** 94.4%, 100.2%, 104.5%, 97.4%, 102.1%

Replicates are samples of about the same size that are carried through an analysis in *exactly* the same way.

1.1 Percentage of stated content



***Student 1*:** 99.5%, 99.9%, 100.2%, 99.4%, 100.5%

***Student 2*:** 95.6%, 96.1%, 95.2%, 95.1%, 96.1%

***Student 3*:** 93.5%, 98.3%, 92.5%, 102.5%, 97.6%

***Student 4*:** 94.4%, 100.2%, 104.5%, 97.4%, 102.1%

Question:

- Why replicate measurements have different result?
- Which measurement is precise and accurate?

Errors in chemical analyses

- It is ***impossible*** to perform a chemical analysis that is ***totally free of errors or uncertainties***.
- We can only hope to ***minimize*** these errors and ***estimate*** their size with ***acceptable accuracy***. Measurement data can only give us an estimate of the "true" value.
  - The concentration of HCl is 0.1025 mol/L (with 0.1% error)

Errors in chemical analyses

- Or, define limits within which the true value of a measured quantity lies with a given level of probability.
  - There is 95% probabilities that copper content is within the range of 0.24 mg to 0.28 mg.

1. Some important terms

- Central value – Mean and Median
- Precision
- Accuracy
- Types of Errors in Experimental Data

Question:

- Why replicate measurements have different result?
- Which measurement is precise and accurate?

1.2 Mean and Median



Six times replicate measurements

Results

19.4 ppm

19.5 ppm

19.6 ppm

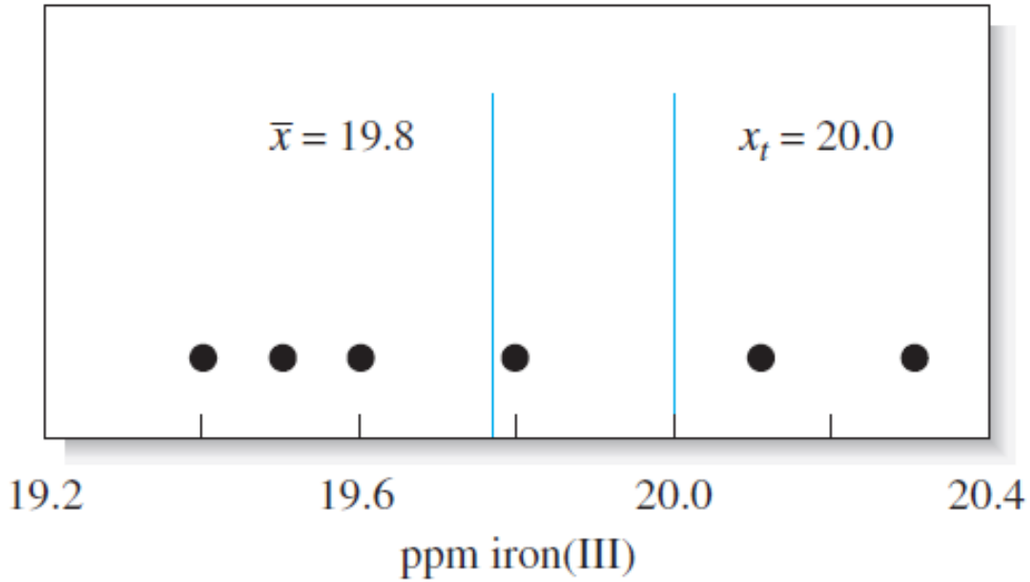
19.8 ppm

20.1 ppm

20.3 ppm

Central Value

Mean **Median**



1.2 Mean and Median



Results

19.4 ppm

19.5 ppm

19.6 ppm

19.8 ppm

20.1 ppm

20.3 ppm

Central Value

$$\text{Mean} = \frac{19.4 + 19.5 + 19.6 + 19.8 + 20.1 + 20.3}{6}$$

$$= 19.78 \approx 19.8 \text{ ppm Fe}$$

- **Mean value**
  - A mean value is obtained by **dividing** the **sum** of a set of replicate measurements by the **number** of individual results in the set.
  - This mean value is also called **arithmetic mean** or **average**.

1.2 Mean and Median



Results

19.4 ppm

19.5 ppm

19.6 ppm

19.8 ppm

20.1 ppm

20.3 ppm

Central Value

$$\textbf{Median} = \frac{19.6 + 19.8}{2} = 19.7 \text{ ppm Fe}$$

- **The median** is the **middle value** in a set of data that has been arranged in numerical. When a set of data has an even number of values, then the median is the **average of the middle pair**.
- 12, 12.3, 12.5, 12.4, 12.1

1.3 Precision

The **closeness of results** to others obtained in exactly the same way

Precision → **Deviation**

Mean = 19.8 ppm Fe

Deviation from the mean $d_i = |x_i - \bar{x}|$



19.4 ppm $d_1 = |19.4 - \mathbf{19.8}| = 0.4$ ppm Fe

19.5 ppm $d_2 = |19.5 - \mathbf{19.8}| = 0.3$ ppm Fe

19.6 ppm $d_3 = |19.6 - \mathbf{19.8}| = 0.2$ ppm Fe

19.8 ppm $d_4 = |19.8 - \mathbf{19.8}| = 0.0$ ppm Fe

20.1 ppm $d_5 = |20.1 - \mathbf{19.8}| = 0.3$ ppm Fe

20.3 ppm $d_6 = |20.3 - \mathbf{19.8}| = 0.5$ ppm Fe

1.3 Precision

The **closeness of results** ← **Precision** → **Deviation** **Mean = 19.8 ppm Fe**

to others obtained in

exactly the same way



19.4 ppm

19.5 ppm

19.6 ppm

19.8 ppm

20.1 ppm

20.3 ppm

Average Deviation $(\overline{d_i})$

$$= \frac{d_1 + d_2 + d_3 + d_4 + d_5 + d_6}{6}$$

$$= \frac{0.4 + 0.3 + 0.2 + 0.0 + 0.3 + 0.5}{6}$$

$$= \mathbf{0.3\ ppm}\ Fe$$

1.3 Precision

The **closeness of results** ← **Precision** → **Deviation** **Mean = 194.8 ppm Fe**

to others obtained in

exactly the same way



194.4 ppm

194.5 ppm

194.6 ppm

194.8 ppm

195.1 ppm

195.3 ppm

Average Deviation $(\overline{d_i})$

$$= \frac{d_1 + d_2 + d_3 + d_4 + d_5 + d_6}{6}$$

$$= \frac{0.4 + 0.3 + 0.2 + 0.0 + 0.3 + 0.5}{6}$$

$$= 0.3\ ppm\ Fe$$

1.3 Precision

The **closeness of results** ← **Precision** → **Deviation** **Mean = 19.8 ppm Fe**

to others obtained in

exactly the same way



19.4 ppm

19.5 ppm

19.6 ppm

19.8 ppm

20.1 ppm

20.3 ppm

Relative Average Deviation

$$= \frac{\bar{d}_i}{\text{mean}} \times 100\% = \frac{0.3}{19.8} = 2\%$$

1.3 Precision

The **closeness of results** ← **Precision** → **Deviation** **Mean = 194.8 ppm Fe**

to others obtained in

exactly the same way



194.4 ppm

194.5 ppm

194.6 ppm

194.8 ppm

195.1 ppm

195.3 ppm

Relative Average Deviation

$$= \frac{\overline{d_i}}{\text{mean}} \times 100\% = \frac{0.3}{194.8} \times 100\%$$

$$= 0.154\% \approx 0.16\%$$

1.3 Precision

The **closeness of results** ← **Precision** → **Deviation** **Mean = 19.8 ppm Fe**

to others obtained in

exactly the same way



19.4 ppm

19.5 ppm

19.6 ppm

19.8 ppm

20.1 ppm

20.3 ppm

Standard Deviation (SD)

$$= \sqrt{\frac{d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2}{6 - 1}}$$

$$= \mathbf{0.35}$$

1.3 Precision

- These are **deviations** of two groups of data. **Which group has better precision?**
  - Group 1： 0.3, 0.2, 0.4, 0.2, 0.1, 0.4, 0.0, 0.3, 0.2, 0.3
  - Group 2： 0.0, 0.1, 0.7, 0.2, 0.1, 0.2, 0.5, 0.2, 0.3, 0.1

$$\text{Group1}\ \bar{d} = \frac{\sum_{i=1}^{10}\left|x_i - \bar{x}\right|}{10} = 0.24 \qquad S_1 = \sqrt{\frac{\sum_{i=1}^{10}(x_i - \bar{x})^2}{10-1}} = 0.28$$

$$\text{Group2}\ \bar{d} = 0.24 \qquad S_2 = 0.33$$

- **Group 1 has better precision.**

- Smaller deviation indicate a better precision.
- However a precise value need not be accurate.

1.3 Precision

The **closeness of results** ← **Precision** → **Deviation** to others obtained in exactly the same way

Mean = 19.8 ppm Fe



19.4 ppm

19.5 ppm

19.6 ppm

19.8 ppm

20.1 ppm

20.3 ppm

Relative Standard Deviation (RSD)

$$= \frac{SD}{mean} \times 100\%$$

$$= \frac{0.35}{19.8} = 1.8\%$$

1.3 Precision

Deviation from the mean $d_i = |x_i - \bar{x}|$

$$\textbf{Average Deviation}(\overline{d_i}) = \frac{d_1 + d_2 + d_3 + d_4 + d_5 + d_6}{6} = \textbf{0.3 ppm Fe}$$

$$\textbf{Relative Average Deviation} = \frac{\bar{d}_i}{\text{mean}} \times 100\% = \frac{0.3}{19.8} = 2\%$$

$$\textbf{Standard Deviation (SD)} = \sqrt{\frac{d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2}{6 - 1}} = \textbf{0.35}$$

$$\textbf{Relative Standard Deviation (RSD)} = \frac{SD}{mean} \times 100\% = \frac{0.35}{19.8} = 1.8\%$$

1.4 Accuracy

- **Accuracy** indicates the closeness of the **measurement** to the **true or accepted value** and is expressed by the **error**.

$$E = x - \mu$$

Error — Measurement — True or accepted value

- Error is the **difference** between the **measurement** and **the true or accepted** value.

About “True value” or “accepted value”

Standards or reference materials

- Reliable experimental method
- In different labs
- By different analysts
- Replicate measures of same sample



© Kresimir Juraga

Standard reference materials from NIST.

1.4 Accuracy

$$E = \bar{x} - \mu$$

- Error is the difference between the measurement and the true or accepted value.

Mean ($\bar{x}$) = 19.8 ppm Fe

True value = 20.0 ppm Fe

$$E = \bar{x} - \mu = 19.8 - 20.0 = -0.2 \text{ ppm Fe}$$

Absolute Error

1.4 Accuracy

- The relative error denotes the **percentage** of **absolute error** compared to the **expected value**.

Relative Error

Mean = 19.8 ppm Fe

True value = 20.0 ppm Fe

$$RE = \frac{E}{\mu} = \frac{\bar{x} - \mu}{\mu} \times 100\% = \frac{19.8 - 20.0}{20.0} \times 100\% = -1\%$$

- ***RE*** **is more useful than** ***E*** **when comparing measurement errors of different quantities.**

1.4 Accuracy

- What are the **relative errors (*RE*)** when we need to weight 2 samples of **0.0021** g and **0.5432** g by using an analytical balance with the **absolute error** of **0.1 mg**?



$$RE = \frac{0.0002}{0.0021} \times 100\% = 9\%$$

$$RE = \frac{0.0002}{0.5432} \times 100\% = 0.04\%$$

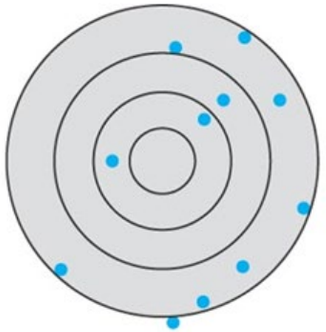
1.4 Accuracy

Question: If the relative error needs to be controlled within **0.1%** in our analysis, *how many grams* we need to weigh at least by using an analytical balance with the ***absolute error of 0.1 mg***?

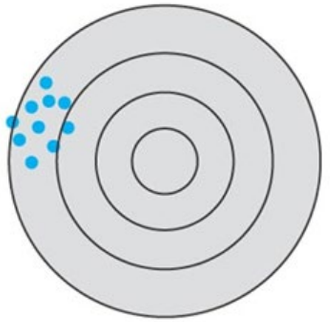




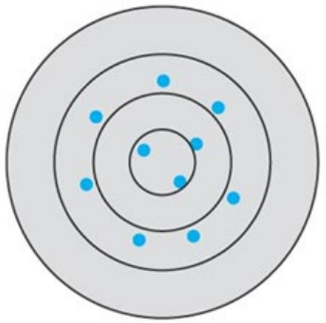
1.4 Precision and Accuracy



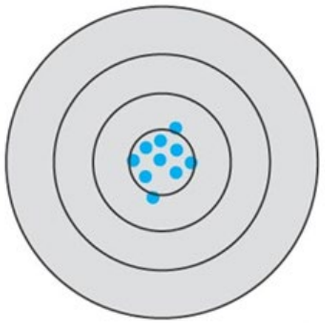
Low accuracy, low precision



Low accuracy, high precision



High accuracy, low precision



High accuracy, high precision

We only consider the accuracy of a set of data when the precision is good.

Example

Student 1: 99.5%, 99.9%, 100.2%, 99.4%, 100.5%

Student 2: 95.6%, 96.1%, 95.2%, 95.1%, 96.1%

Student 3: 93.5%, 98.3%, 92.5%, 102.5%, 97.6%

Student 4: 94.4%, 100.2%, 104.5%, 97.4%, 102.1%

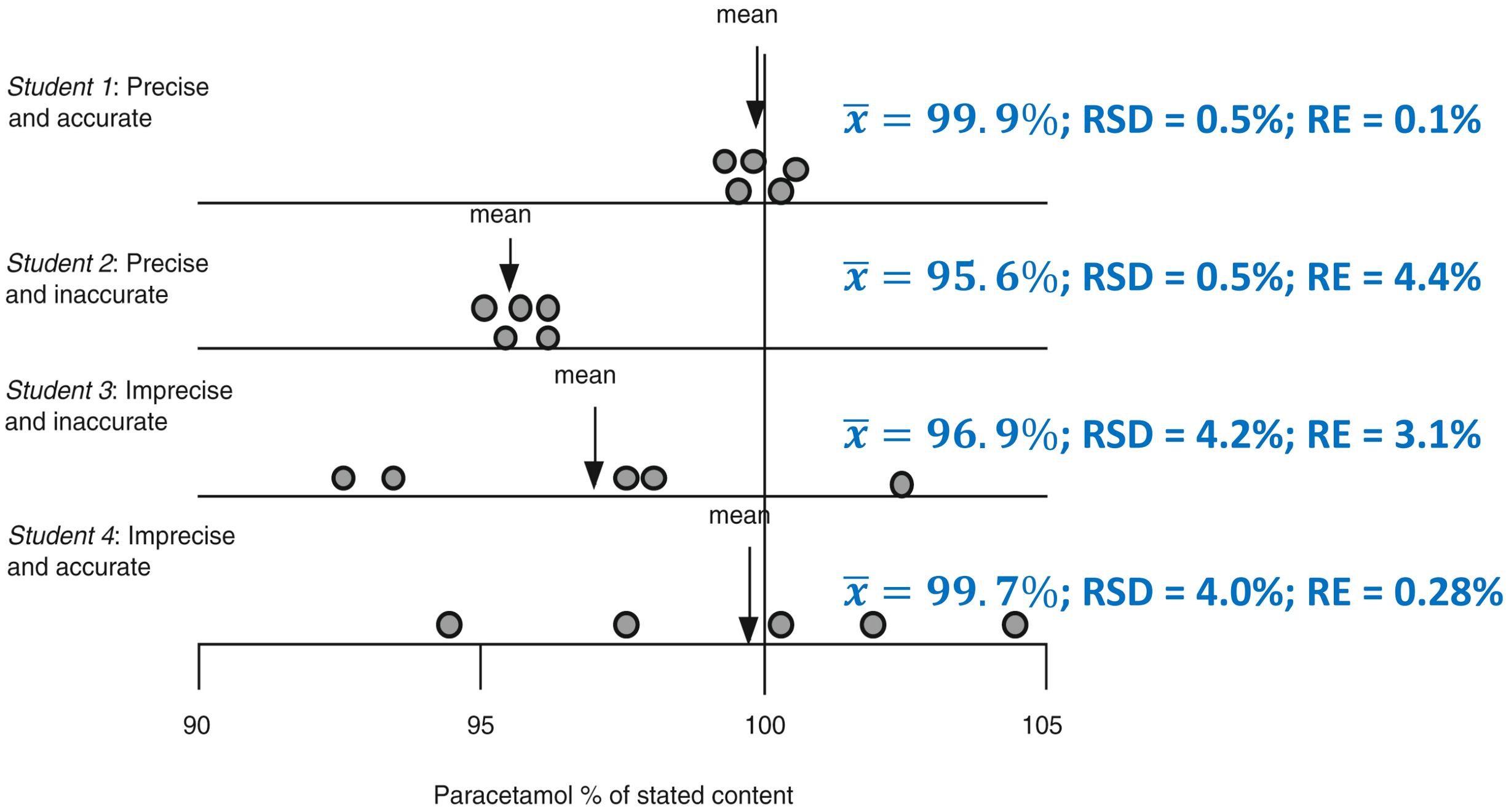


Student 1: $\bar{x} = 99.9\%$; **RSD = 0.5%; RE = 0.1%** **Precise and accurate**

Student 2: $\bar{x} = 95.6\%$; **RSD = 0.5%; RE = 4.4%** **Precise and less accurate**

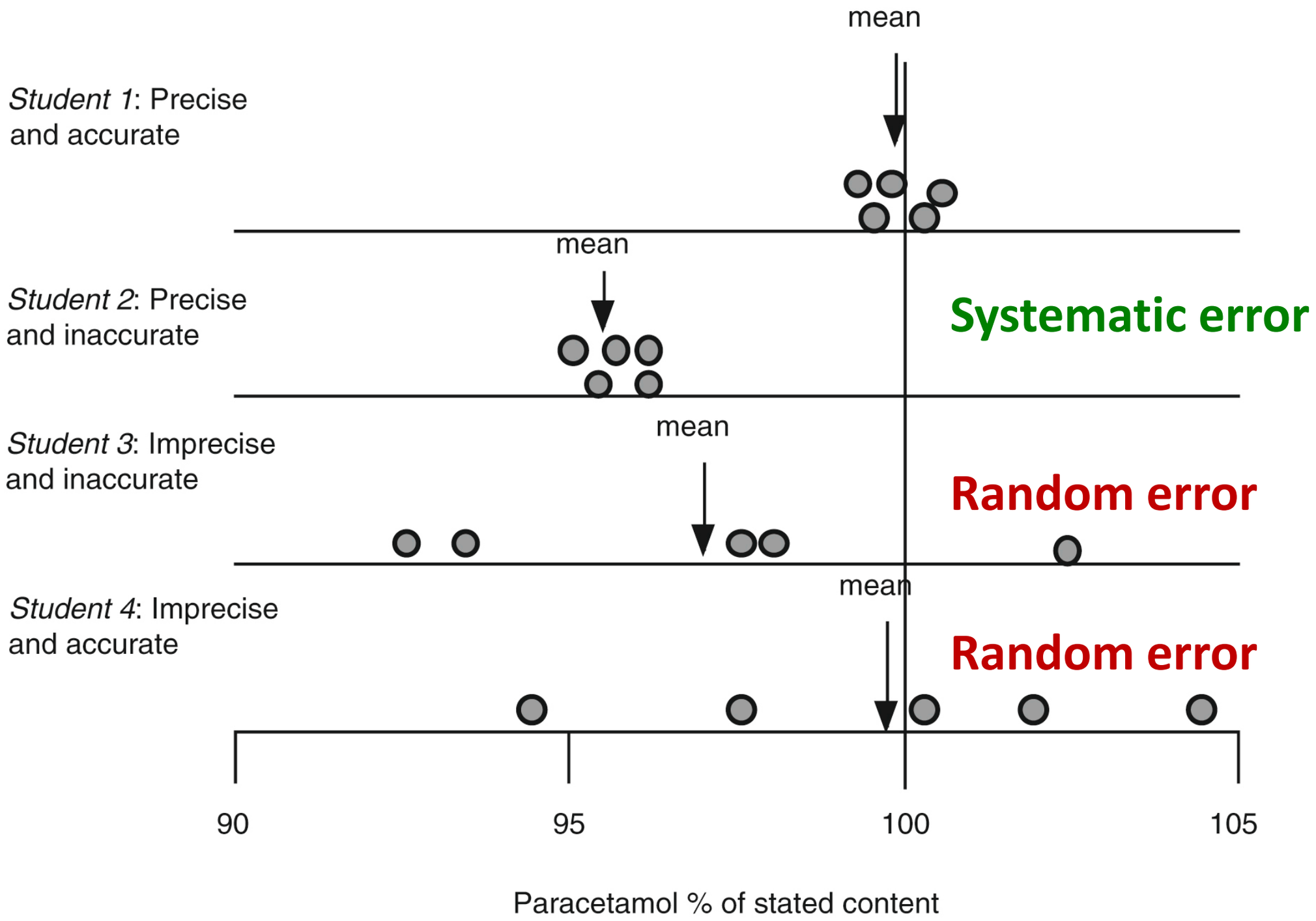
Student 3: $\bar{x} = 96.9\%$; **RSD = 4.2%; RE = 3.1%** **Imprecise and less accurate**

Student 4: $\bar{x} = 99.7\%$; **RSD = 4.0%; RE = 0.28%** **Imprecise**


mean
Student 1: Precise and accurate
$\bar{x} = 99.9\%$; RSD = 0.5%; RE = 0.1%
mean
Student 2: Precise and inaccurate
$\bar{x} = 95.6\%$; RSD = 0.5%; RE = 4.4%
mean
Student 3: Imprecise and inaccurate
$\bar{x} = 96.9\%$; RSD = 4.2%; RE = 3.1%
mean
Student 4: Imprecise and accurate
$\bar{x} = 99.7\%$; RSD = 4.0%; RE = 0.28%
90
95
100
105
Paracetamol % of stated content

1.5 Types of Errors in Experimental Data

- **Two main types**
  - **Determinate errors / Systematic errors**
    - the mean of a set of data to **differ** from the accepted value.
    - It affects the **accuracy** of results.
  - **Indeterminate errors / or Random errors**
    - data to be **scattered** more or less symmetrically around a **mean value**.
    - It determines the reproducibility of measurements and therefore their **precision**.
    - Scatter increase, precision decrease.


mean
Student 1: Precise and accurate
mean
Student 2: Precise and inaccurate
Systematic error
mean
Student 3: Imprecise and inaccurate
Random error
mean
Student 4: Imprecise and accurate
Random error
90
95
100
105
Paracetamol % of stated content

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01 Some important terms

02 Systematic Errors in Chemical Analysis

03 Random Errors in Chemical Analysis

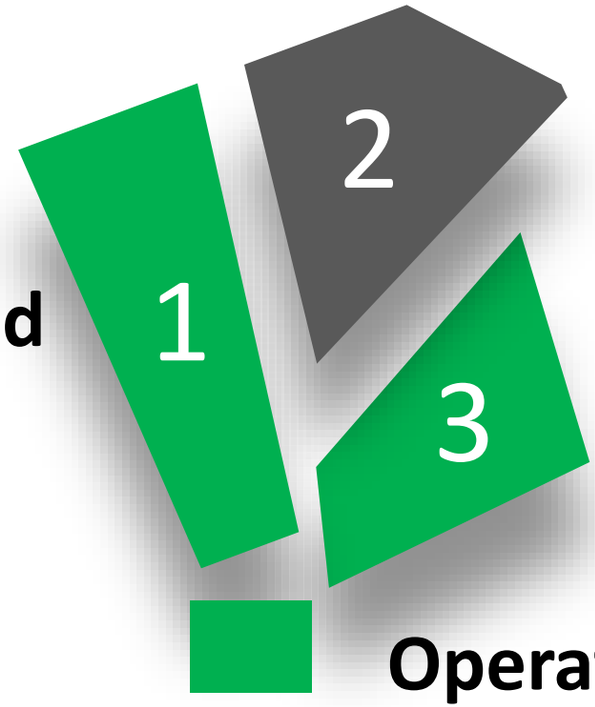
04 Statistical Data Treatment and Evaluation

2. Systematic Errors / Determinate error

- **Definition：** it is an error caused by ***certain fixed factors*** during the analytical process.
- **Characteristics:**
  - It has a ***fixed direction*** (positive or negative) and magnitude.
  - Its magnitude and sign ***can be determined*** and ***can be corrected or eliminated***.
  - It reappears upon **repeated** measurements.
  - It can be **detected** by changing experimental conditions.

2.1 Source of Systematic Errors

These errors are determinable and can be **avoided** if care is taken.



Instrumental and reagent error

Method errors

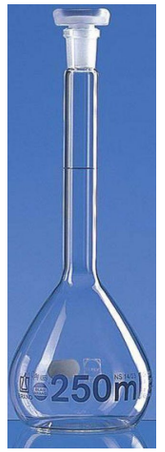
Operational and personal error

Instrumental and reagent error

- These errors are caused by non-ideal instrument behavior, by ***faulty calibrations***, or by use under inappropriate condition.
- Sometimes an instrument error may arise from the ***environmental factors*** on the instrument.
  - — a pipette calibrated at 20°C, if used at 30°C will introduce error in volume.
- Instrumental errors may largely be eliminated by periodically ***calibrating*** the instruments.

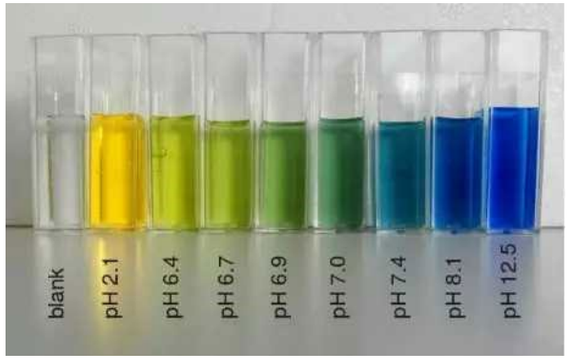






Method error

- These errors are caused by adopting defective experimental methods.
  - In volumetric analysis the use of an ***improper indicator*** leading to wrong results is an example for methodic error.

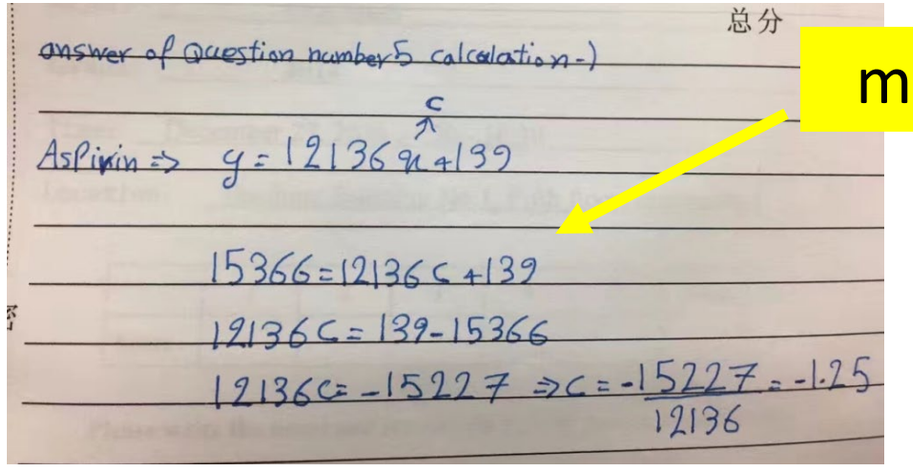


- Proper understanding of the ***theoretical background*** of the experiments is a necessity for avoiding methodic errors.

Operational and personal error

- This type errors is introduced because of variation of **personal judgments**.
- For example due to color blindness a person may arrive at wrong results in a volumetric or colorimetric analysis.
- **Using incorrect mathematical equations** and committing **arithmetic mistakes** will also cause operative errors.





2.3 Detection of Systematic Instrument and Personal Errors

Some systematic instrument errors can be found and corrected by calibration.

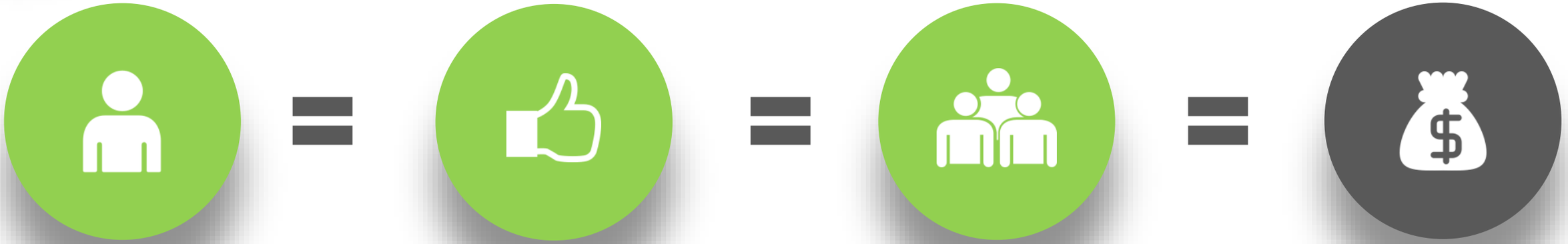




2.3 Detection of Systematic Instrument and Personal Errors

- ***Periodic calibration of equipment*** because the response of most instruments changes with time as a result of component aging, corrosion, or mistreatment.
- Most personal errors can be minimized by **careful, disciplined laboratory work**. It's a good habit to check instrument readings, notebook entries, and calculations systematically.
- Errors due to **limitation of the experimenter** can usually be avoided by carefully choosing the ***analytical method*** or using an ***automated procedure***.

2.4 Detection of Systematic Method Errors



Analyzing Standard Samples

Using an Independent Analytical Method

Performing Blank Determinations

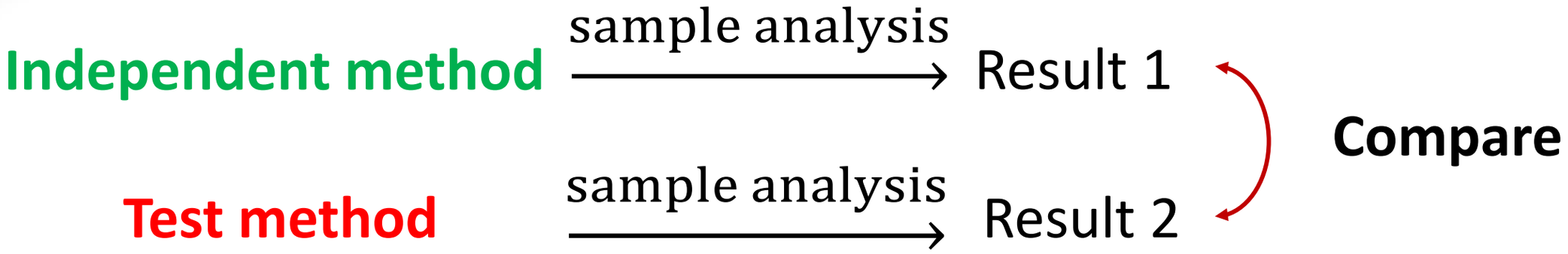
Varying the Sample Size

Analyzing Standard Samples

- Analyzing ***standard reference materials*, *SRM***, is the **best way** to estimate the bias of an analytical method.
- The SRM contain one or more analytes at well-known or certified concentration levels.
- Standard materials can sometimes be prepared by **synthesis**.
- Standard reference material can be **purchased** from a number of ***governmental*** and ***industrial sources***.



Using an Independent Analytical Method



The independent method should differ as much as possible from the one under study to minimize the possibility that some **common factor** in the sample **has the same effect** on both methods.

Performing Blank Determinations

- In a blank, all steps of the analysis are performed in the **absence of a analyte**.
- The results from the blank are then applied as a **correction** to the sample measurements.
- Reveal errors due to interfering contaminants from the reagents and vessels employed in analysis.

$$\text{Sample} \xrightarrow{\text{analysis}} \text{Result 1}$$

$$\text{Blank} \xrightarrow{\text{analysis}} \text{Result 2}$$

without analyte

A blank determination is an analysis on the added reagents only.

Varying the Sample Size

- As the **size** of a measurement **increases**, the effect of a constant **error decreases**.
- Thus, constant errors can often be detected by varying the sample size.



Question: If the relative error needs to be controlled within 0.1% in our analysis, how many grams we need to weigh at least by using an analytical balance with the absolute error of 0.1 mg?

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01 Some important terms

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3.1 The Nature of Random Errors

- These errors are also called **accidental errors** / **indeterminate errors**.
- Indeterminate errors arise from **uncertainties** in a measurement.

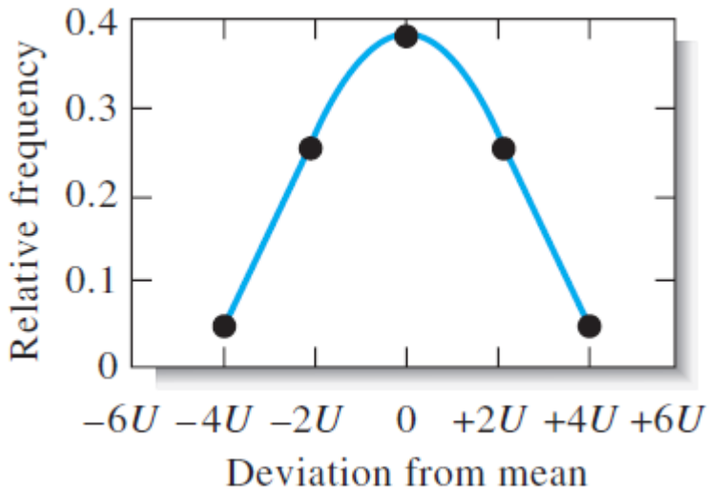
Cannot be controlled

Nonreproducibly by the experimentalist

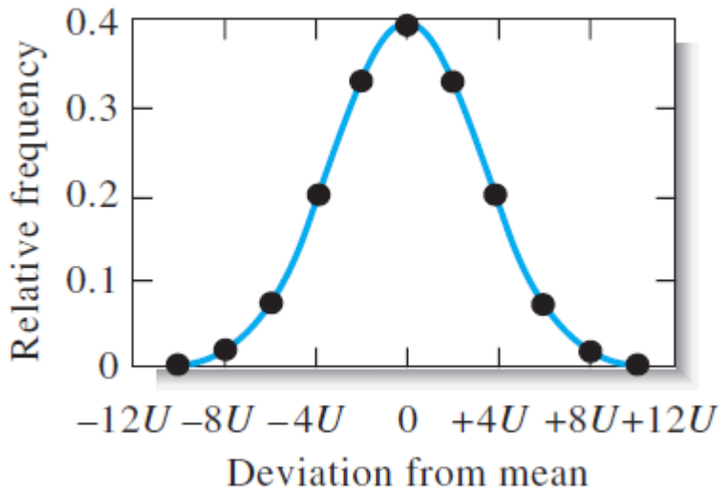
Cannot be totally eliminated

3.1 The Nature of Random Errors

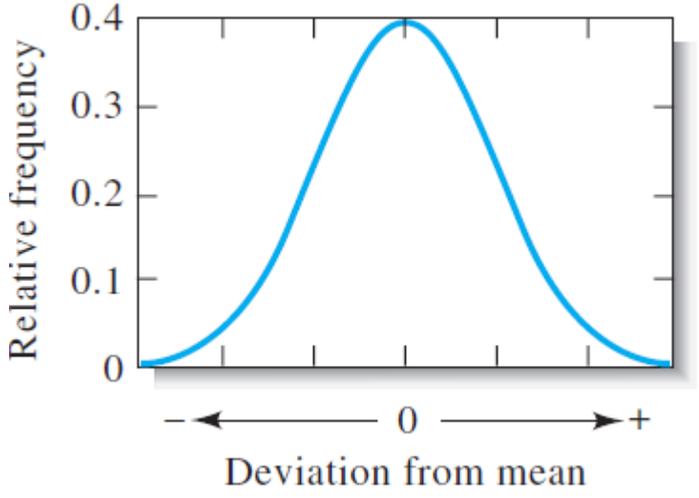
Distribution of Experimental Results



(a) Four random uncertainties



(b) ten random uncertainties



(c) a very large number of random uncertainties.

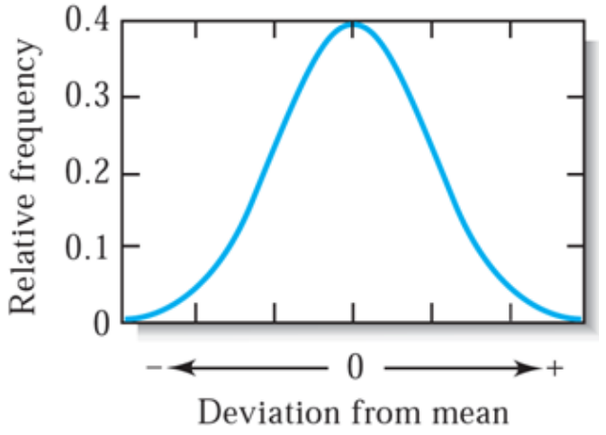
Frequency distribution for measurements

Indeterminate errors should follow a normal distribution, or Gaussian curve.

3.1 The Nature of Random Errors

Distribution of Experimental Results

- An **equal number** and type of ***positive*** and ***negative*** errors occur;
- **Small errors** occur much ***more frequently*** than large errors;
- **Very large errors** occur very ***infrequently***;
- The value that occurs ***most often*** is the one with no error, namely, the **true value**.



A. Samples and Populations

- The **population** is the collection of **all measurements** of interest and must be carefully defined by the experimenter.
  - In some cases, the population is **finite and real**, while in others, the population is **hypothetical or conceptual in nature**.
- **Sample** is the collection of **limited measurements**.

Statistical Sampling Principles

3.2 Statistical Treatment of Random Errors

A. Samples and Populations



Calcium



B. The Population Mean μ and the Sample Mean $\bar{x}$

- The **sample mean $\bar{x}$** is the arithmetic average of a **limited sample** drawn from **a population of data**.
- The **population mean μ**, in contrast, is the **true mean** for the population.
- In the **absence of systematic error**, the **population mean μ is also the true value** for the measured quantity.
- Particularly when ***N* is small**, x differs from μ because a small sample of data may not exactly represent its population.

B. The Population Mean μ and the Sample Mean $\bar{x}$

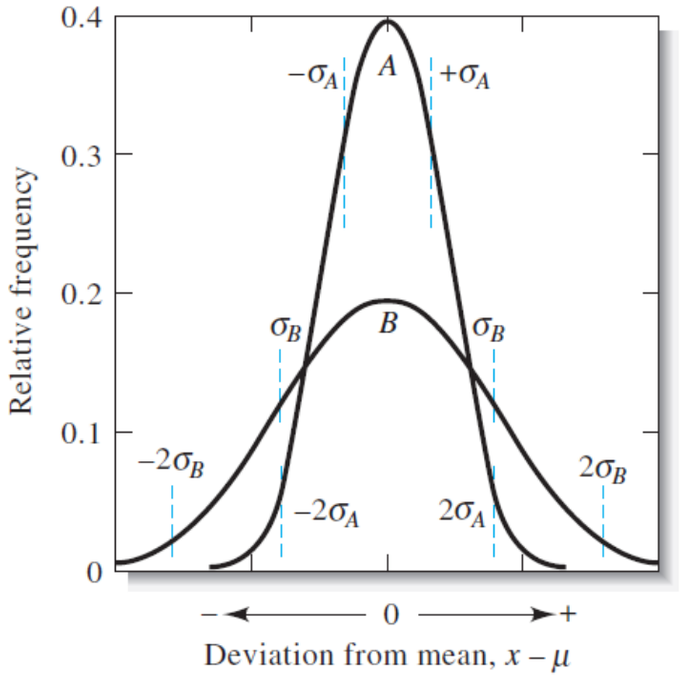
- In most cases we do not know μ and **must infer its value from $\bar{x}$**.
- The probable difference between x and μ decreases rapidly as the number of measurements making up the sample increases. Usually by the time ***N* reaches 20 to 30**, this **difference is negligible**.
- Note that the sample mean $\bar{x}$ is a statistic that **estimates** the population parameter μ.

C. The Population Standard Deviation σ

The **population standard deviation** σ, which is a measure of the precision of the population

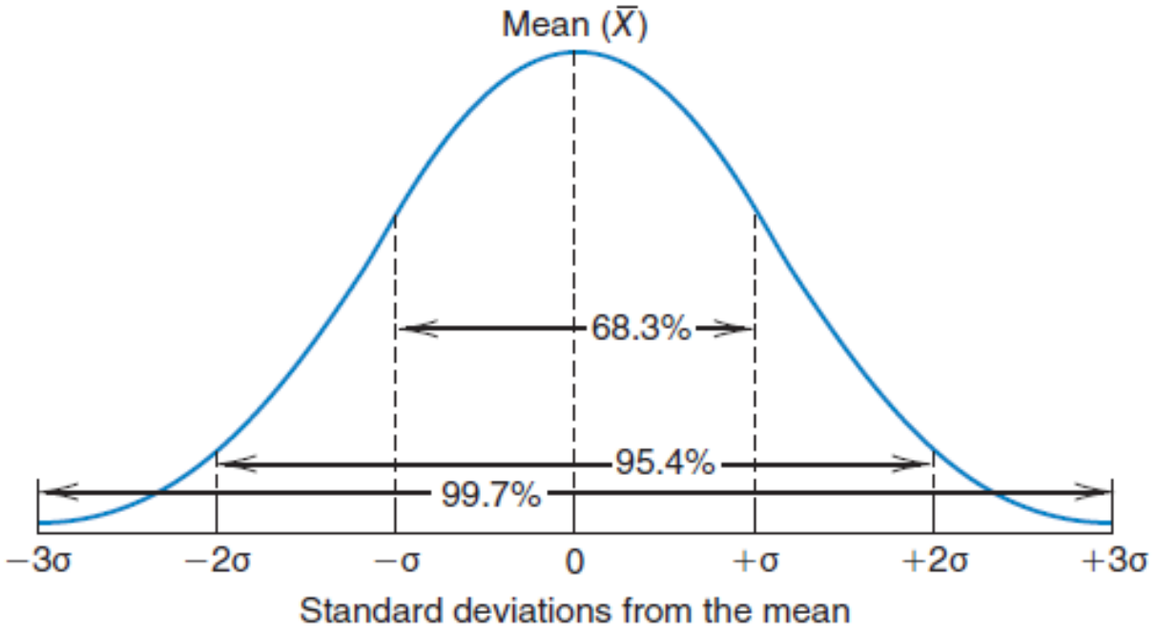
$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}}$$

The precision of the data set leading to curve *A* is twice as good as that of the data set represented by curve *B*.



3.2 Statistical Treatment of Random Errors

C. The Population Standard Deviation σ



$$z = \frac{x - \mu}{\sigma}$$

3.2 Statistical Treatment of Random Errors

D. The Sample Standard Deviation – *s*

$$s = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \bar{x})^2}{N-1}} = \sqrt{\frac{\sum_{i=1}^{N} d_i^2}{N-1}}$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}}$$

number of degrees of freedom (*N* – 1)

Chp.2

01 Some important terms

02 Systematic Errors in Chemical Analysis

03 Random Errors in Chemical Analysis

04 Statistical Data Treatment and Evaluation

4. Statistical Data Treatment and Evaluation

- Confidence Intervals
- Tests of Significance
- Detection of Gross Errors
- Significant Figure



4.1 Confidence Intervals

A huge number of measurements (approaching infinity) would be required

Impossible

Limited number of measurements

Estimate

- **The Population Mean μ and the Sample Mean $\bar{x}$**

The Population Standard Deviation σ

The Sample Standard Deviation s

- For example, we might say that it is 99% probable that the true population mean μ for a set of potassium measurements lies in the interval $7.25 \pm 0.15\%$ K.
- Thus, the probability that the mean lies in the interval from 7.10 to 7.40% *K* is 99%.

4.1 Confidence Intervals

Confidence limit

- For example, we might say that it is **99% probable** that the true population mean μ for a set of potassium measurements lies in the **interval** 7.25 $\pm$ 0.15% K.

- Thus, the probability that the mean lies in the interval from **7.10 to 7.40%** K is 99%.

Confidence interval

- **Confidence interval:** It is an interval surrounding the experimentally determined mean $\bar{x}$ within which the population mean μ is expected to lie with a **certain degree of probability**.

- **Confidence limits:** the limits of the interval.

4.1 Confidence Intervals

When we use sample mean $\bar{x}$ and standard deviation ***s*** to estimate the confidence intervals, we define ***t*** as:

$$t = \frac{|\bar{x} - \mu|}{S}\sqrt{N}$$

$$\mu = \bar{x} \pm \frac{tS}{\sqrt{N}}$$

Values of *t* for Various Levels of Probability

Degrees of Freedom	80%	90%	95%	99%	99.9%
1	3.08	6.31	12.7	63.7	637
2	1.89	2.92	4.30	9.92	31.6
3	1.64	2.35	3.18	5.84	12.9
4	1.53	2.13	2.78	4.60	8.61
5	1.48	2.02	2.57	4.03	6.87
6	1.44	1.94	2.45	3.71	5.96
7	1.42	1.90	2.36	3.50	5.41
8	1.40	1.86	2.31	3.36	5.04
9	1.38	1.83	2.26	3.25	4.78
10	1.37	1.81	2.23	3.17	4.59
15	1.34	1.75	2.13	2.95	4.07
20	1.32	1.73	2.09	2.84	3.85
40	1.30	1.68	2.02	2.70	3.55
60	1.30	1.67	2.00	2.62	3.46
∞	1.28	1.64	1.96	2.58	3.29

Degrees of Freedom $f = N - 1$

4.1 Confidence Intervals

A clinical chemist obtained the following data for the alcohol content of a sample of blood:

- % of C_2H_5OH: **0.084, 0.089, and 0.079.**
- Calculate the 95% confidence interval for the mean, assuming that the three results obtained are the only indication of the precision of the method

$$\bar{x} = \frac{0.084 + 0.089 + 0.079}{3} = 0.084\%\ C_2H_5OH$$

$$s = \sqrt{\frac{(0.084 - 0.084)^2 + (0.084 - 0.089)^2 + (0.084 - 0.079)^2}{3 - 1}} = 0.005\%\ C_2H_5OH$$

4.1 Confidence Intervals

A clinical chemist obtained the following data for the alcohol content of a sample of blood:

- % of C_2H_5OH: **0.084, 0.089, and 0.079.**
- Calculate the 95% confidence interval for the mean, assuming that the three results obtained are the only indication c

$$\mu = \bar{x} \pm \frac{tS}{\sqrt{N}}$$

$f = N - 1 = 3 - 1 = 2$

$t = 4.30$

Values of *t* for Various Levels of Probability

Degrees of Freedom	80%	90%	95%	99%	99.9%
1	3.08	6.31	12.7	63.7	637
2	1.89	2.92	4.30	9.92	31.6
3	1.64	2.35	3.18	5.84	12.9
4	1.53	2.13	2.78	4.60	8.61
5	1.48	2.02	2.57	4.03	6.87
6	1.44	1.94	2.45	3.71	5.96
7	1.42	1.90	2.36	3.50	5.41
8	1.40	1.86	2.31	3.36	5.04
9	1.38	1.83	2.26	3.25	4.78
10	1.37	1.81	2.23	3.17	4.59
15	1.34	1.75	2.13	2.95	4.07
20	1.32	1.73	2.09	2.84	3.85
40	1.30	1.68	2.02	2.70	3.55
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4.1 Confidence Intervals

A clinical chemist obtained the following data for the alcohol content of a sample of blood:

- % of C_2H_5OH: **0.084, 0.089, and 0.079.**
- Calculate the 95% confidence interval for the mean, assuming that the three results obtained are the only indication of the precision of the method

$t = 4.30$ $\quad \bar{x} = 0.084\%\ C_2H_5OH$ $\quad s = 0.005\%\ C_2H_5OH$

$$\mu = \bar{x} \pm \frac{ts}{\sqrt{N}}$$

$$\mu = 0.084 \pm \frac{4.30 \times 0.005}{\sqrt{3}} = 0.084 \pm 0.012\% C_2H_5OH$$

$0.084 - 0.012 =$ **0.072 % C_2H_5OH** $\xleftrightarrow[95\%]{\mu}$ $0.084 + 0.012 =$ **0.096 % C_2H_5OH**

Practice

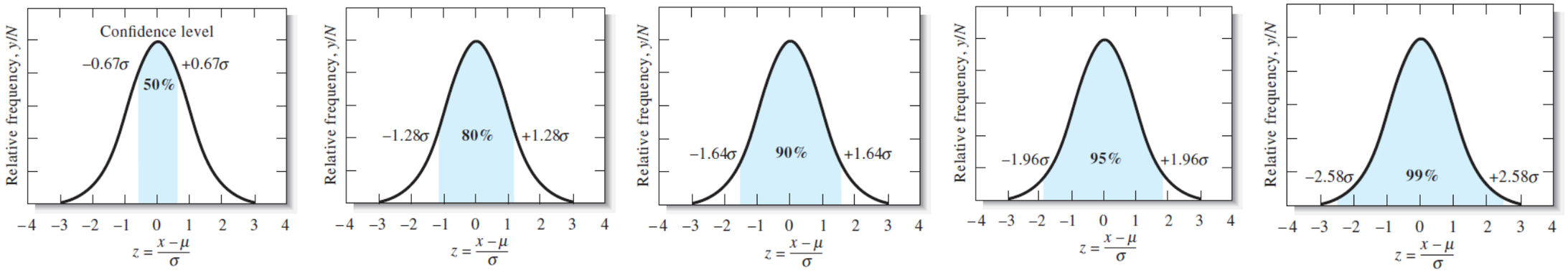
Using the information given in each row of the table below, determine the item marked with a question mark (?).

The true mean (μ) is the mean with the confidence interval at the confidence level indicated as determined by the Student's *t* method, for example 27.48 ± 0.14.

	Mean	n – 1	Confidence Level (%)	s	t (from Table 1.1)	μ
(a)	16.37	5	90	0.28	?	?
(b)	0.238	6	95	0.019	?	?

4.1 Confidence Intervals

- **Confident level (*P*)**
  - Is the probability that the **true mean** lies within a certain interval and is often expressed as a **percentage**.
- **Significance level (*α*)**
  - The probability that a result is **outside** the confidence interval.
  - α = 1 - P



4.2 Tests of significance

- Is the **mean** of a data set significantly **different** from the **true value**?
- Are the **means of two different data** sets **significantly different**?
- Are the **precisions** of two different data sets **significantly different**?

4.2 Tests of significance

Question 1: Is the mean $\bar{x}$ of a data set significantly different from the true value μ?

Determinate errors ⟶ **Accuracy**

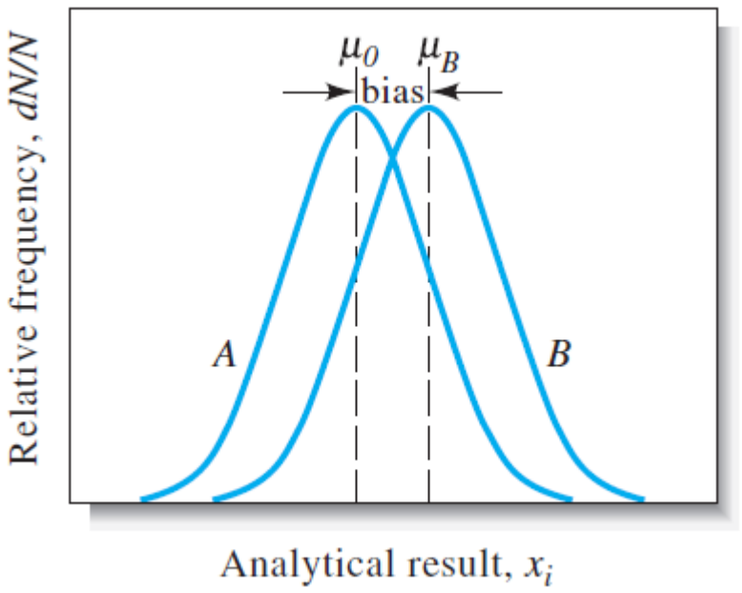
μ_0 is a specific value of μ such as an **accepted value**, a theoretical value, or a threshold value.

For a small number of results, we use $\bar{x}$ to estimate μ

$$\mu = \bar{x} \pm \frac{ts}{\sqrt{N}}$$

4.2 Tests of significance

Question 1: Is the mean $\bar{x}$ of a data set significantly different from the true value *m*?



Method A: has **no systematic error**, or bias, random errors would give the frequency distribution shown by curve A.

$\mu_A = \mu_0$ **no bias**

Method B: has **some systematic error** so that $\overline{x_B}$, which estimates m_B, differs from the accepted value μ_0.

Bias = $\mu_B - \mu_0$

The *t* test is used to determine the significance of the difference.

Is the mean $\bar{x}$ of a data set significantly different from the true value μ?

- **A new procedure** for determining trace amounts of zinc in vegetables was evaluated by using it to determine the zinc content of a standard sample, with the following results:
  - **0.083 μg/g**, **0.088 μg/g**, **0.087 μg/g** and **0.086 μg/g** Zn.
  - The **certified value of the standard sample** is **0.082 μg/g** Zn.
  - Is there a **significant difference** between the mean of the results and the certified value?

μ_0 = 0.082 mg/g Zn

Solution:

$$\bar{x} = \frac{0.083 + 0.088 + 0.087 + 0.086}{4} = \mathbf{0.086\ \mu g/g\ Zn}$$

$$s = \sqrt{\frac{0.003^2 + 0.002^2 + 0.001^2 + 0.000^2}{4 - 1}} = 0.0022\ \mu g/g$$

Is the **mean** $\bar{x}$ of a data set significantly **different** from the **true value** μ?

- **A new procedure** for determining trace amounts of zinc in vegetables was evaluated by using it to determine the zinc content of a standard sample, with the following results:
  - **0.083 μg/g**, **0.088 μg/g**, **0.087 μg/g** and **0.086 μg/g** Zn.
  - The **certified value of the standard sample** is **0.082 μg/g** Zn.
  - Is there a **significant difference** between the mean of the results and the certified value?

Solution:

$$\mu = \bar{x} \pm \frac{ts}{\sqrt{N}}$$

If $\mu = \mu_0$, then

$$t = \frac{|\bar{x} - \mu_0|}{s}\sqrt{N} = \frac{|0.086 - 0.082|}{0.0022}\sqrt{4} = 3.64$$

Is the mean $\bar{x}$ of a data set significantly different from the true value μ?

Values of *t* for Various Levels of Probability

Degrees of Freedom	80%	90%	95%	99%	99.9%
1	3.08	6.31	12.7	63.7	637
2	1.89	2.92	4.30	9.92	31.6
3	1.64	2.35	3.18	5.84	12.9
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10	1.37	1.81	2.23	3.17	4.59
15	1.34	1.75	2.13	2.95	4.07
20	1.32	1.73	2.09	2.84	3.85
40	1.30	1.68	2.02	2.70	3.55
60	1.30	1.67	2.00	2.62	3.46
∞	1.28	1.64	1.96	2.58	3.29

$$t = 3.64$$

$$\mu = \bar{x} \pm \frac{ts}{\sqrt{N}}$$

At confidence level of **95%**:

$$t_{\alpha,f} = 3.18 < 3.64$$

There is a ***significant difference*** between the mean of the results and the certified or "true" value when the **confidence level is 95%.**

Is the mean $\bar{x}$ of a data set significantly different from the true value μ?

- If

$$t = \frac{|\bar{x} - \mu_0|}{s}\sqrt{N} > t_{\alpha,f}$$

  - We may assume that the determinate error is significant.

- If

$$t = \frac{|\bar{x} - \mu_0|}{s}\sqrt{N} < t_{\alpha,f}$$

  - we may assume that the determinate error is NOT significant.

4.2 Tests of significance

- Is the **mean** of a data set significantly **different** from the **true value**?
- Are the **means of two different data** sets **significantly different**?
- Are the **precisions** of two different data sets **significantly different**?

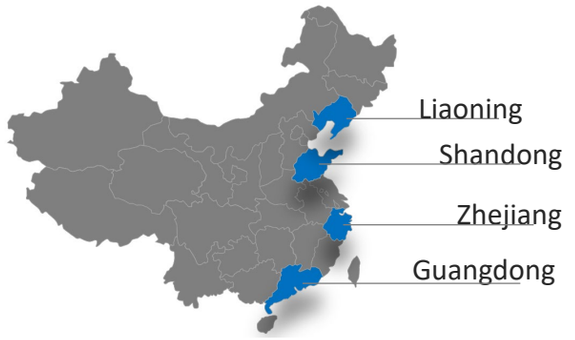
4.2 Tests of significance

Question 2: Are the means of two different data sets significantly different?

- In many cases, it needs to be determined whether a difference in the means of two sets of data is **real or the result of random error**.
- In some cases, the results of chemical analyses are used to determine whether two **materials** are identical.









Question 2: Are the means of two different data sets significantly different?



Six times **replicate** measurements

% of alcohol content

$\bar{x} = 12.61\%$



Is the wine in the glass from the bottle?







Four times **replicate** measurements

$\bar{x} = 12.53\%$



Question 2: Are the means of two different data sets significantly different?

% of alcohol content



Six times **replicate** measurements

$\bar{x} = 12.61\%$ ⟶ $\mu_1 = \overline{x_1} \pm \frac{ts_1}{\sqrt{N_1}}$



Is the wine in the glass from the bottle?



Four times **replicate** measurements

$\bar{x} = 12.53\%$ ⟶ $\mu_2 = \overline{x_2} \pm \frac{ts_2}{\sqrt{N_2}}$

Question 2: Are the means of two different data sets significantly different?

$$\textbf{If, } \mu_1 = \mu_2 \rightarrow \overline{x_1} \pm \frac{ts_1}{\sqrt{N_1}} = \overline{x_2} \pm \frac{ts_2}{\sqrt{N_2}} \rightarrow t = \frac{\overline{x_1} - \overline{x_2}}{s_{pooled}\sqrt{\frac{N_1 + N_2}{N_1 N_2}}}$$

$$s_{pooled} = \sqrt{\frac{(N_1 - 1)s_1^2 + (N_2 - 1)s_2^2}{N_1 + N_2 - 2}}$$

Question 2: Are the means of two different data sets significantly different?



Six times **replicate** measurements

% of alcohol content

$\overline{x} = 12.61\%$

$$t = \frac{\overline{x_1} - \overline{x_2}}{s_{pooled}\sqrt{\frac{N_1 + N_2}{N_1 N_2}}}$$





Four times **replicate** measurements

$\overline{x} = 12.53\%$

$$t = \frac{12.61 - 12.53}{s_{pooled}\sqrt{\frac{6 + 4}{6 \times 4}}} = 1.77$$

s_{pooled} = 0.070%

Question 2: Are the means of two different data sets significantly different?

Values of t for Various Levels of Probability

Degrees of Freedom	80%	90%	95%	99%	99.9%
1	3.08	6.31	12.7	63.7	637
2	1.89	2.92	4.30	9.92	31.6
3	1.64	2.35	3.18	5.84	12.9
4	1.53	2.13	2.78	4.60	8.61
5	1.48	2.02	2.57	4.03	6.87
6	1.44	1.94	2.45	3.71	5.96
7	1.42	1.90	2.36	3.50	5.41
8	1.40	1.86	2.31	3.36	5.04
9	1.38	1.83	2.26	3.25	4.78
10	1.37	1.81	2.23	3.17	4.59
15	1.34	1.75	2.13	2.95	4.07
20	1.32	1.73	2.09	2.84	3.85
40	1.30	1.68	2.02	2.70	3.55
60	1.30	1.67	2.00	2.62	3.46
∞	1.28	1.64	1.96	2.58	3.29

How to find the degree of freedom?

$$f = N_1 + N_2 - 2$$

$$f = 6 + 4 - 2 = 8$$

$$t_{\alpha,f} = t_{0.05,8} = \mathbf{2.31}$$

$$t = \mathbf{1.77}$$

Since **1.771 < 2.31**, we accept at the 95% confidence level and conclude that there is **no difference** in the alcohol content of the wines.

Question 2: Are the means of two different data sets significantly different?

- In other cases, the results are used to determine whether two analytical **methods** give the same values or whether two **analysts** using the same methods obtain the same means.

Analyst 1



$$\xrightarrow[\text{Same sample}]{\text{same method}} \overline{x_1}$$

Analyst 2



$$\xrightarrow[\text{Same sample}]{\text{same method}} \overline{x_2}$$

Expert



Same sample

$$\xrightarrow{Method\ 1} \overline{x_1}$$

$$\xrightarrow{Method\ 2} \overline{x_2}$$

4.2 Tests of significance

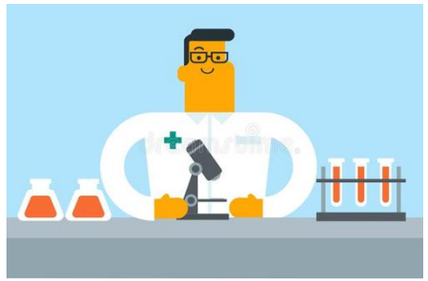
Is the **mean** of a data set significantly **different** from the **true value**?

Are the **means of two different data** sets **significantly different**?

Are the **precisions** of two different data sets **significantly different**?

Question 3: Are the precisions of two different data sets significantly different?

Analysis are done by using the same sample, same procedure, reagents, and instruments.



Senior technician

1.38 %
1.33 %
1.34 %
1.35 %
1.30 %



New Assistant

1.28 %
1.36 %
1.35 %
1.40 %
1.31 %

- Is the results new assistant obtained as good as the senior technician?
- Is there any significantly different in **precision** between these two groups of data?

$s_{senior} = 0.029\ \%$

$s_{new} = 0.046\ \%$

Standard Deviation, *s*

Is the difference in precisions (*s*) significant?

Question 3: Are the precisions of two different data sets significantly different?

F test

We define:

s^2 as variance, present as **V**

$$s^2_{senior} = (0.029\ \%)^2$$

$$s^2_{new} = (0.046\ \%)^2$$

$$F_c = \frac{s^2_{new}}{s^2_{senior}} = \frac{V_{new}}{V_{senior}} = \frac{(0.046\ \%)^2}{(0.029\ \%)^2} = 2.5$$

Note: In *F* test, the **larger** variance is always the **numerator**.

Question 3: Are the precisions of two different data sets significantly different?

N = 5



1.38 %
1.33 %
1.34 %
1.35 %
1.30 %

N = 5



1.28 %
1.36 %
1.35 %
1.40 %
1.31 %

$$F_c = \frac{s^2_{new}}{s^2_{senior}} = \frac{V_{new}}{V_{senior}} = \frac{(0.046\ \%)^2}{(0.029\ \%)^2} = \mathbf{2.5}$$

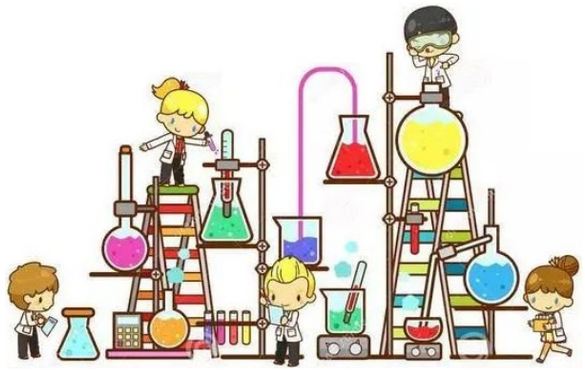
Critical Values of F at the 5% Probability Level (95 % confidence level)

Degrees of Freedom (Denominator)	Degrees of Freedom (Numerator)								
	2	3	4	5	6	10	12	20	∞
2	19.00	19.16	19.25	19.30	19.33	19.40	19.41	19.45	19.50
3	9.55	9.28	9.12	9.01	8.94	8.79	8.74	8.66	8.53
4	6.94	6.59	6.39	6.26	6.16	5.96	5.91	5.80	5.63
5	5.79	5.41	5.19	5.05	4.95	4.74	4.68	4.56	4.36
6	5.14	4.76	4.53	4.39	4.28	4.06	4.00	3.87	3.67
10	4.10	3.71	3.48	3.33	3.22	2.98	2.91	2.77	2.54
12	3.89	3.49	3.26	3.11	3.00	2.75	2.69	2.54	2.30
20	3.49	3.10	2.87	2.71	2.60	2.35	2.28	2.12	1.84
∞	3.00	2.60	2.37	2.21	2.10	1.83	1.75	1.57	1.00

The tabular value F_t **is 6.39**. Since F_c **= 2.5** does not exceed this value, the hypothesis of no difference is **accepted**.

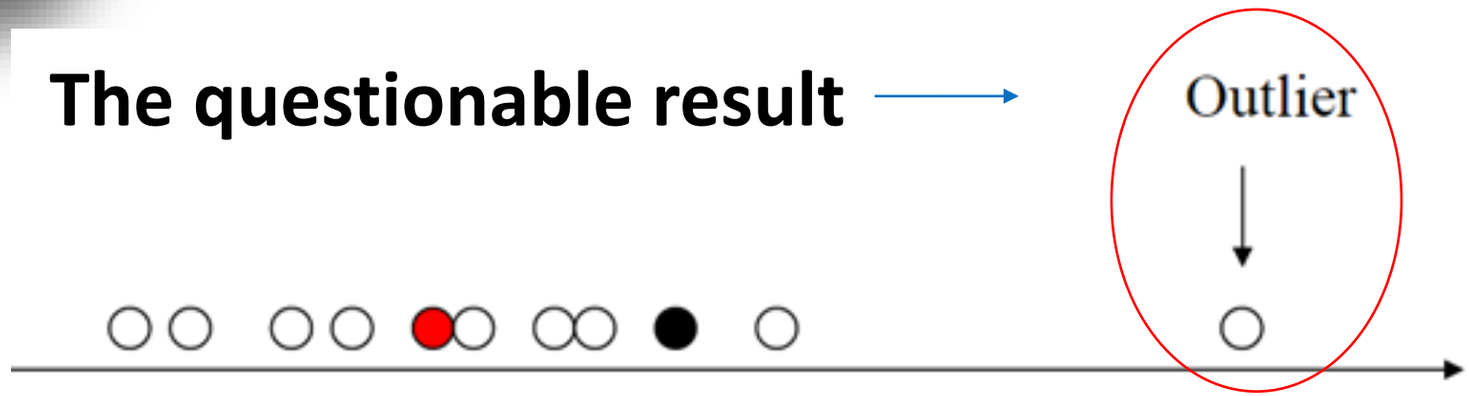
Question 3: Are the precisions of two different data sets significantly different?

- The F test may be used to provide information concerning:
  - (a) whether Method A is **more precise** than Method B
  - (b) whether there is **a difference in the precision** of two methods
- If the calculated $F_c \geq F_t$, there is **significant difference** in precision of two different data.



4.3 Detection of Gross Errors

The questionable result →



Retain or Reject?



An **outlier** is a result that is ***quite different*** from the others in the data set and might be due to a **gross error**.

They are often the product of human errors.

Q test

4.3 Detection of Gross Errors – Q test

x_q – **questionable** result x_n – **nearest neighbor** result



Calcite sample

N = 5

Retain or Reject?

$$Q = \frac{|x_q - x_n|}{w}$$

w – spread of the entire set

% CaO	55.95	56.00	56.04	56.08	56.23
				x_n	x_q

w = 56.23 – 55.95 = 0.28

$$Q = \frac{|56.23 - 56.08|}{0.28} = 0.54$$

4.3 Detection of Gross Errors – Q test

Critical Values for the Rejection Quotient, Q^*

Number of Observations	Q_{crit} (Reject if $Q > Q_{crit}$) 90% Confidence	95% Confidence	99% Confidence
3	0.941	0.970	0.994
4	0.765	0.829	0.926
5	0.642	0.710	0.821
6	0.560	0.625	0.740
7	0.507	0.568	0.680
8	0.468	0.526	0.634
9	0.437	0.493	0.598
10	0.412	0.466	0.568

$$Q = \frac{|56.23 - 56.08|}{0.28} = 0.54$$

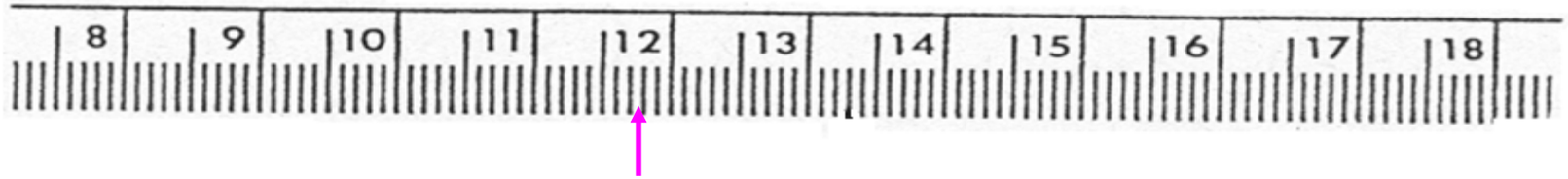
N = 5

For five measurements, Q_{crit} at the **90% confidence level** is 0.64.

Since 0.54 < 0.64, we must retain 56.23.

4.4 Significant figures – Definition

- Scientists use significant figures to **determine how precise** a measurement is.
- Significant digits in a measurement include
  - **all of the known digits** + one **estimated** digit



11.7**?** cm 11.7**6** cm

Estimation

~~16.3~~ cm? → 16.3**0** cm? √

Estimation

Each line is 0.1cm

Zero here is your estimation of the last digit which ***can not be ignored*** in data recording.

4.4 Significant figures – Definition



- When weighing an sample on a balance, the measurement fluctuates in the final decimal place.
- We record this sample's mass as 0.5730 g $\pm$ 0.0001 g.

4.4 Significant figures – Definition

Absolute Error (*E*)



± 0.1 g



± 0.01 g



± 0.001 g

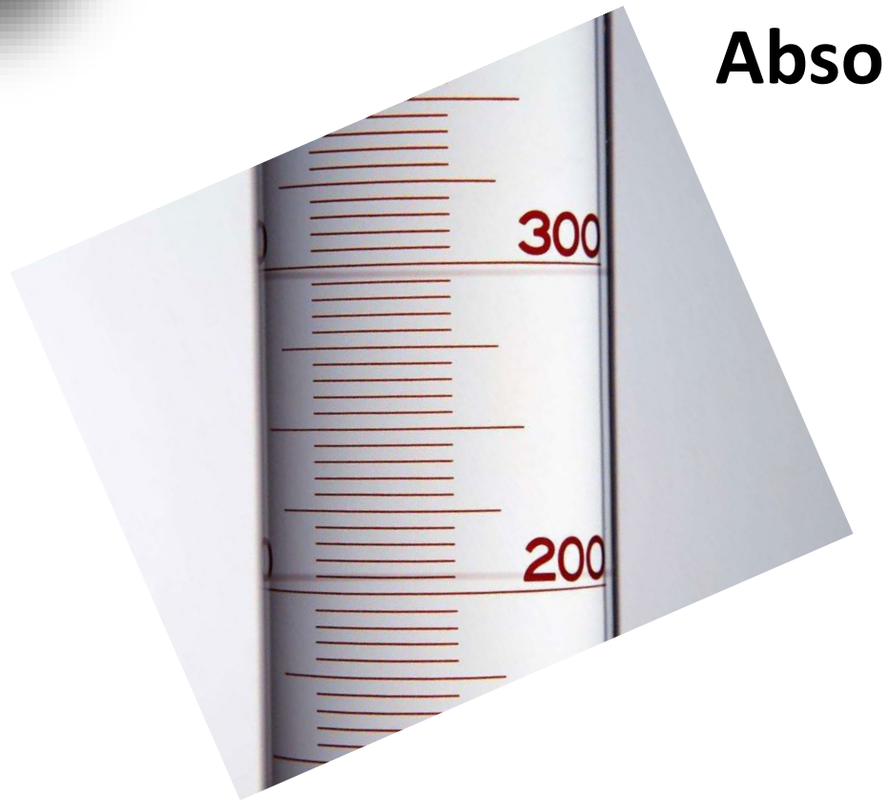


± 0.0001 g



± 0.00001 g

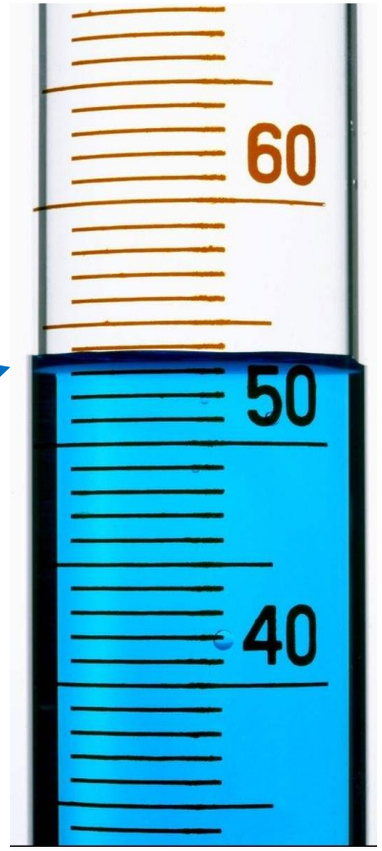
4.4 Significant figures – Definition

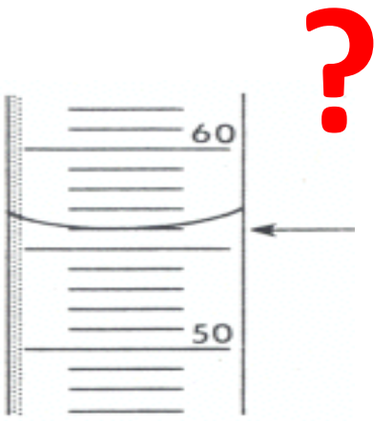


Absolute Error (E)

0.1 ml

53.5 ml





All of the certain digits + the first uncertain digit

4.4 Significant figures – Numbers of Significant Figures

- **Rule 1.**
  - **All NON zero digits are ALWAYS significant**
  - Example
    - 1234 g, 56.87 ml, 1.2654 g, 5.4 mmol/L
- **Rule 2.**
  - **About Zeroes**
    - Leading zeros (place holders)
    - Captive zeros
    - Trailing zeros

4.4 Significant figures – Numbers of Significant Figures

- **Leading zeros (place holders)**
  - The first significant figure in a measurement is the first digit other than zero counting from left to right.
  - Example: 0.0045g
    - (4 is the 1st sig. fig.)
    - “0.00” are place holders.
    - The zeros are not significant

4.4 Significant figures – Numbers of Significant Figures

- **Captive zeros**
  - Zeros within a number at **always significant**
  - Example 30.0809 g

    All digits are significant
- How many significant digits are in the following numbers?

504	3 Significant Figures
60002	5 Significant Digits
9.077	4 Significant Figures

4.4 Significant figures – Numbers of Significant Figures

- **Trailing zeros**
  - At the **end of numbers** but to the right of the **decimal point**.
  - 2.00 g
    - has 3 sig. digits (what this means is that the measuring instrument can measure exactly to two decimal places.
  - When 2.00g reports as 2000mg, the situation becomes confusing.
    - use scientific notation and report the mass as 2.0×10^3 mg

  Zeros are significant if a number contains **decimals.**

4.4 Significant figures – Numbers of Significant Figures

- How many significant digits are in the following numbers?

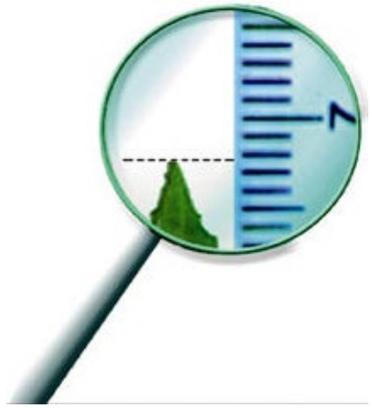
32.0	3 Significant Figures
19.000	5 Significant Digits
105.0020	7 Significant Figures

4.4 Significant figures – Numbers of Significant Figures

- Another way to say this is:
  - zeros are only significant if
    - They are **between** significant digits OR
    - They are the very final thing at the end of a decimal.

4.4 Significant figures – Numbers of Significant Figures

- Example
  - How many significant digits are in the following numbers?



0.0012	2 Significant Digit
6.002×10^{23}	4 Significant Digits
300.00	5 Significant Digits
230000	confusing
500	confusing

4.4 Significant figures – Rounding significant figures

202.99	(want 3 SF)	203
16.32	(want 2 SF)	16
265.50	(want 3 SF)	266
3.5450	(want 3 SF)	3.54
0.0599	(want 1 SF)	0.06
74.59	(want 2 SF)	75

- If the digit to the immediate right of the last significant digit is **greater that a 5**, you round up the last significant figure.
- If the digit to the immediate right of the last significant digit is **less that 5**, do not round up the last significant digit.
- If the number to the immediate right of the last significant is a 5, and that **5 is followed by a zero**, you look at the **last significant digit and make it even**.
- If the number to the immediate right of the last significant is a 5, and that **5 is followed by a non zero digit**, **round up**

4.4 Significant figures – Scientific Notation

- **Very large numbers**
  - If the number you start with is greater than 1, the exponent will be **positive**.
  - Write the number **39923** in scientific notation
    - First move the decimal until 1 number is in front – **3.9923**
    - Now at × 10 – **3.9923 × 10**
    - Now count the number of decimal places that you moved **(4)**
    - Since the number you started with was greater than 1, the exponent will be **positive**.
    - **3.9923×10^{4}**

4.4 Significant figures – Scientific Notation

- **Very small numbers**
  - If the number you start with is less than 1, the exponent will be **negative**
  - Write the number **0.0052** in scientific notation
  - First move the decimal until 1 number is in front **– 5.2**
  - Now at × 10 – **5.2 × 10**
  - Now count the number of decimal places that you moved (3)
  - Since the number you started with was less than 1, the exponent will be **negative**
  - **5.2×10^{-3}**

4.4 Significant figures – Scientific Notation

- Place the following numbers in scientific notation

455.87	4.5587×10^{2}
6800.1	6.8001×10^{3}
0.00863	8.63×10^{-3}
0.07856	7.856×10^{-2}
578158.1	5.781581×10^{5}



4.4 Significant figures – Significant figures in calculation

Addition and Subtraction



Multiplication and Division

Logarithms and Antilogarithms

4.4 Significant figures – Significant figures in calculation

Addition and Subtraction

- How many significant figures need to be kept in the final calculation depends on the number which has minimum number after the decimal.
- Example 3.4 + 0.020 + 7.31

  = 10.73

  = 10.7

  0.5362 + 0.001 + **0.25**

  = 0.7872

  =0.79
- All numbers being added or subtracted can be rounded to the least significant unit ***before*** adding or subtracting.

Addition and Subtraction

- Calculate the sum of 8.5784, 5.6, 0.078, 5.89.

8.5784 + 5.6 + 0.078 + 5.89

= 8.58 + 5.6 + 0.08 + 5.89

= 20.15

= 20.2

Multiplication and Division

- How many significant figures need to be kept in the final calculation depends on the number which has **the smallest number of significant digits**.

$$\frac{24 \times 4.52}{100.0} = 1.08 \qquad answer = 1.1$$

$$\frac{24 \times 4.02}{100.0} = 0.965 \qquad answer = 0.96$$

Logarithms and Antilogarithms

- Example: lg (**4.000** × 10^{-5}) = -4.397940008672038

  = - 4.**3979**

  - In a **logarithm** of a number, keep as many digits to the **right of the decimal point** as there are **significant figures in the original number.**

- Example: antilg (12.**5**) = 3.16227766 ×10^{12}

  = **3** ×10^{12}

  - In an **antilogarithm** of a number, keep as many digits as there are digits to **the right of the decimal point in the original number**.

Summary

01 Some important terms

- Precision
- Accuracy
- Mean and Median
- Types of Errors in Experimental Data

02 Systematic Errors in Chemical Analysis

- Source of Systematic Errors
- Detection of Systematic Errors

03 Random Errors in Chemical Analysis

- The nature of random errors
- Statistical Treatment of Random Errors

04 Statistical Data Treatment and Evaluation

Summary

Statistical Data Treatment and Evaluation (Part 1 and 2)

Confidence Intervals

- Confidence level (P)
- Significance level (α)
- Confidence interval
- Confidence limit
- Population mean μ and sample mean $\bar{x}$
- Population deviation σ and sample deviation s
- How to use sample mean $\bar{x}$ to estimate the population mean μ by t test

Tests of Significance

$$\mu = \bar{x} \pm \frac{ts}{\sqrt{N}}$$

- Is the mean of a data set significantly different from the true value?
- Are the means of two different data sets significantly different?
- Are the precisions of two different data sets significantly different?

Summary

Statistical Data Treatment and Evaluation (Part 3 and 4)

Detection of Gross Errors

- Q test

$$Q = \frac{|x_q - x_n|}{w}$$

Significant Figures

- Definition
- Numbers of Significant Figures
- Rounding significant figures
- Scientific Notation
- Significant figures in calculation

The end of Chapter 2